Hierarchical Shell Manifold Theory: A Graded Operator Framework, Relational Benchmarks, and the Physical-Selection Problem

Vincent Mark Garrett
Independent Researcher, Lubbock, Texas, USA
vince1976.garrett@outlook.com

October 2026
Revised framework manuscript

Abstract

Hierarchical Shell Manifold Theory (HSMT) investigates graded operator structures and the circumstances under which projected sectors can support autonomous or approximately autonomous descriptions. We give a self-contained minimal formulation using a Hilbert space, orthogonal grade projections, and a reference operator. We prove the elementary operator-theoretic consequences under explicit domain assumptions, distinguish discrete grades from measured dimension, and formulate exact and approximate sector persistence. A continuous spectral parent can reproduce grades by coarse-graining but does not select a physically preferred grading. As a finite relational benchmark, we construct contingency-table rectangle moves, the associated graph Laplacian, its three-vertex example, and distinct reversible dynamics with different nonzero frequency ratios. We also state a precisely delimited positivity obstruction to obtaining an indefinite quadratic form by positive block Schur complementation. The broader HSMT research program reports finite hierarchy constructions, scalable relational phases, and continuum geometry; these are presented here as claims requiring their full defining data and proofs in technical companions, rather than as independently proved theorems of this article. We identify a nonuniqueness boundary: kinematic grading, exact reduction, positive-norm conservation, and relational geometry do not alone specify an operational time, a physical generator, a measurement map, or an observed spacetime dimension. The contribution of the present article is a transparent mathematical foundation, a reproducible benchmark, and a falsifiable ledger separating proved results, conditional constructions, program claims, and physical questions.

Keywords: graded Hilbert spaces; Feshbach reduction; emergent hierarchy; relational geometry; graph dynamics; physical selection; foundations of physics.

1 Scope and claim standard

The distinction between mathematical possibility and physical identification is central to theories of emergence. A self-adjoint operator is not, by itself, an experimentally selected Hamiltonian; a graph metric is not automatically spatial distance; an index labeling direct-sum sectors is not necessarily an observed spatial dimension. HSMT is intended as a framework for keeping these identifications separate.

This paper has two purposes. First, it supplies a small, independently checkable mathematical foundation and a finite relational benchmark. Second, it records the status of more ambitious hierarchy-generation and geometric constructions without representing their unprovided proofs as established here. We distinguish four types of assertion throughout:

Established mathematical fact
A standard result cited from the literature or proved below.

Conditional construction
A result whose validity depends on a stated choice of graph, operator, dynamics, or other input.

Program claim
A result reported by the HSMT development program for which this article does not reproduce enough definitions and proof to certify it independently.

Open physical identification
A proposed connection to empirical physics for which no selection or measurement correspondence has yet been supplied.

This convention is not cosmetic: no claim of independent mathematical verification is made here for the program claims.

2 The graded operator framework

Definition 2.1 (Minimal graded core)**.** *A minimal graded core is a triple* $(\mathcal{H}, \{P_N\}_{N\in\mathcal{I}}, D)$*, where* $\mathcal{H}$ *is a separable complex Hilbert space,* $\mathcal{I}$ *is finite or countable,* P_N *are mutually orthogonal projections,*

$$P_N P_M = \delta_{NM} P_N, \qquad \sum_{N\in\mathcal{I}} P_N = \mathrm{I} \quad \textit{in the strong operator topology,} \tag{1}$$

and D *is a densely defined linear operator on* $\mathcal{H}$*. We write* $\mathcal{H}_N = P_N\mathcal{H}$*.*

The datum D is deliberately a *reference operator*, not necessarily the physical time-evolution generator. The definition does not assert that any particular N corresponds to a number of spatial dimensions or that a preferred physical grade exists. If block expressions $P_N D P_M$ are employed for unbounded D, their domains must be specified; they are not automatically operators defined on all of $\mathcal{H}_M$.

Proposition 2.2 (Bounded intergrade coupling)**.** *Suppose* D_0 *is self-adjoint on* $\mathrm{Dom}(D_0)$ *and* $T \in \mathcal{B}(\mathcal{H})$ *is self-adjoint. Then* $D_0 + T$ *is self-adjoint on* $\mathrm{Dom}(D_0)$*.*

Proof. A bounded T is D_0-bounded with relative bound zero. The conclusion is the bounded-perturbation case of the Kato–Rellich theorem [1, 2]. □

This theorem permits controlled intergrade hopping in a chosen model. Neither boundedness nor a nearest-neighbour hopping pattern follows from Eq. (1); both are independent assumptions.

2.1 Exact reduction and memory

Let $\mathcal{H} = \mathcal{H}_P \oplus \mathcal{H}_Q$, where P is an orthogonal projection and $Q = \mathrm{I} - P$. We consider a particularly transparent domain-controlled class:

$$H = \begin{pmatrix} A & B \\ B^* & C \end{pmatrix}, \qquad \mathrm{Dom}(H) = \mathrm{Dom}(A) \oplus \mathrm{Dom}(C), \tag{2}$$

with self-adjoint A, C and bounded $B : \mathcal{H}_Q \to \mathcal{H}_P$. These assumptions make H self-adjoint by Proposition 2.2.

Proposition 2.3 (Projected resolvent)**.** *For $z \in \rho(C)$, let*

$$S_P(z) = z - A - B(z - C)^{-1}B^*, \qquad \mathrm{Dom}(S_P(z)) = \mathrm{Dom}(A). \tag{3}$$

Then $z \in \rho(H)$ if and only if $S_P(z)$ has a bounded, everywhere-defined inverse from $\mathcal{H}_P$ to $\mathrm{Dom}(A)$ *(with its graph norm). In that case,*

$$P(z - H)^{-1}P\big|_{\mathcal{H}_P} = S_P(z)^{-1}. \tag{4}$$

Proof. For $(z - H)(u, v) = (f, g)$ the second block equation gives $v = (z - C)^{-1}(g + B^*u)$. Substitution in the first yields $S_P(z)u = f + B(z - C)^{-1}g$. This triangular block factorization has bounded off-diagonal factors and proves the stated inverse equivalence and identity. The result is a conventional Schur–Feshbach formula [3, 4, 5]. □

For an arbitrary $z \notin \mathbb{R}$, self-adjointness ensures $z \in \rho(C) \cap \rho(H)$, so Eq. (4) applies without additional spectral qualifications.

Writing $\psi(t) = (u(t), v(t))$ and taking $\mathrm{i}\partial_t\psi = H\psi$ gives

$$v(t) = e^{-\mathrm{i}Ct}v(0) - \mathrm{i}\int_0^t e^{-\mathrm{i}C(t-s)}B^*u(s)\,\mathrm{d}s, \tag{5}$$

when the evolution and initial data admit the displayed strong expressions (or in the usual mild sense). Therefore

$$\mathrm{i}\partial_t u(t) = Au(t) + Be^{-\mathrm{i}Ct}v(0) - \mathrm{i}\int_0^t Be^{-\mathrm{i}C(t-s)}B^*u(s)\,\mathrm{d}s. \tag{6}$$

The operator-valued memory factor is $Be^{-\mathrm{i}Ct}B^*$; the $-\mathrm{i}$ prefactor belongs to the integral in this convention. Equations Eqs. (4) and (6) are exact for an *already selected* projection P. They do not select which P is physically observable.

2.2 Sector persistence

For a bounded self-adjoint generator H and an orthogonal projection P, define the leakage operator $L_P = (\mathrm{I} - P)HP$ and $\varepsilon_P = \|L_P\|$. A reducing sector satisfies $L_P = 0$ (equivalently, $[H, P] = 0$ for self-adjoint bounded H). For $t \in \mathbb{R}$,

$$\|(\mathrm{I} - P)e^{-\mathrm{i}tH}P\| \le \min\{1, |t|\varepsilon_P\}. \tag{7}$$

Indeed, the derivative of $(\mathrm{I} - P)e^{-\mathrm{i}tH}P$ can be expressed through the Duhamel formula relative to the block-diagonal operator $PHP + (\mathrm{I} - P)H(\mathrm{I} - P)$, and integration gives the bound. For unbounded H such conclusions require further domain or bounded-commutator hypotheses.

Persistent sectors are thus meaningful after a generator is chosen, not an automatic consequence of grading.

3 Continuous parents and measured dimension

Proposition 3.1 (Coarse-graining a projection-valued measure)**.** *Let E be a projection-valued measure on a measurable space (X, Σ), and let $\{\Delta_N\}_{N\in\mathcal{I}}$ be a countable measurable partition of X up to an E-null set. Then $P_N = E(\Delta_N)$ satisfies Eq.* (1)*.*

Proof. Multiplicativity and countable additivity of the projection-valued measure give orthogonality and strong resolution of the identity. □

This shows that the graded core is compatible with a continuous parent. It does not show that a continuous parent is uniquely determined by a given grading, or that a particular partition is selected in nature.

A physical dimension estimator requires an operationally specified response $Q_N(\ell)$ and a scale range in which it can be measured. One possible definition is

$$d_N(\ell) = \frac{\mathrm{d}\log Q_N(\ell)}{\mathrm{d}\log \ell}, \qquad Q_N(\ell) > 0. \tag{8}$$

The dimension inferred depends on the choice of Q_N: spectral, volume-growth, information, and dynamical dimensions need not agree. In particular, $N = d_N$ is not a theorem. A monotonically relabeled continuous parameter also carries no preferred metric or distance scale until an independent ruler is introduced.

4 Generative hierarchy: definitions and verification boundary

An HSMT research direction introduces *Bounded External Generative Complexity* (BEGC) to constrain the complexity of responses among composite states. In the original research summary, the intended contrast was a shared bounded-complexity response rule versus an arbitrary table of pairwise relations. The source summary does not specify the encoding class, complexity measure, growth limit, or exact descent rules. Consequently, the verbal statement is insufficient to define a mathematical predicate for which a general theorem could be proved here.

A complete technical definition must at least identify: (i) a class $\mathscr{S}_n$ of exposed states; (ii) an admissible family of response maps R_n; (iii) a complexity functional $\mathsf{C}(R_n, \mathscr{S}_n)$; (iv) a precise uniform or asymptotic bound; and (v) the stability and predictive-congruence conditions that make a quotient an autonomous grade. Without these, the assertion that four microstates are necessary and sufficient for a minimal two-grade system cannot be independently evaluated.

4.1 Reported finite and scalable program results

The broader program records the following constructions; they are *not established in this paper*:

(i) A finite, symmetry-neutral construction based on $J(8, 3)$ reportedly selects multiple simultaneous persistence-stable predictive quotients using a three-body rainbow statistic and a collision functional.

(ii) A scalable five-coordinate parent reportedly supports a symmetry-broken $3 + 2$ phase, approximate quotient closure, and finite representatives.

(iii) On associated fingerprint and contingency-switch structures, a path operator, response-locality criterion, several dimensional estimators, and an elliptic continuum limit have been reported.

To elevate these to theorems in a submission, a companion must state its configuration space and functional, prove the asserted minimizers and open phase region, quantify quotient defects, provide convergence hypotheses and limiting topology, and document the numerical or symbolic checks. Until such a companion is accessible, the claims are research targets/results as reported by the program, not independently reproduced contributions of this article.

5 An exact finite relational benchmark

We now give a self-contained construction that does not rely on the preceding program claims.

Definition 5.1 (Contingency fibre and rectangle switch)**.** *For nonnegative integer margins* $r = (r_1, \dots, r_m)$ *and* $c = (c_1, \dots, c_n)$ *with* $\sum_i r_i = \sum_j c_j$*, define*

$$\mathcal{F}(r,c) = \left\{ X \in \mathbb{Z}_{\geq 0}^{m\times n} : \sum_j X_{ij} = r_i, \quad \sum_i X_{ij} = c_j \right\}. \tag{9}$$

A rectangle move changes four cells in distinct rows i, k *and columns* j, l *by* $\pm(E_{ij}+E_{kl}-E_{il}-E_{kj})$ *whenever the resulting table remains nonnegative. The* switch graph *has vertex set* $\mathcal{F}(r,c)$ *and an undirected edge for each admissible move.*

For two-way tables without structural zeros, elementary 2×2 moves connect each nonempty fixed-margin fibre; this is a standard Markov-basis fact, not a new HSMT theorem [6]. Constraints or quotient identifications can change the statement and require separate verification.

Give each undirected edge unit conductance and let A_G be the adjacency matrix and Δ_G the degree matrix. The combinatorial Laplacian

$$K = \Delta_G - A_G \tag{10}$$

is positive semidefinite because

$$\langle f, Kf \rangle = \sum_{\{x,y\}\in E(G)} |f(x) - f(y)|^2 \geq 0. \tag{11}$$

The construction is a relational operator on the state graph; by itself it is not an observed spatial Laplacian.

Example 5.2 (Three-state benchmark)**.** *For* 2×2 *margins* $r = c = (2, 2)$*, every table is*

$$X_a = \begin{pmatrix} a & 2-a \\ 2-a & a \end{pmatrix}, \qquad a \in \{0, 1, 2\}. \tag{12}$$

One admissible switch increments or decrements a *by one. The switch graph is the path* P_3*, with Laplacian*

$$K = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}, \qquad \operatorname{spec}(K) = \{0, 1, 3\}. \tag{13}$$

The eigenvectors $(1,1,1)^T$*,* $(1,0,-1)^T$*, and* $(1,-2,1)^T$ *have the respective eigenvalues* $0, 1, 3$*.*

5.1 Different dynamics on the same relational graph

The static data K permit inequivalent dynamical completions. For $g > 0$, one Schrödinger-type choice is

$$\mathrm{i}\dot{\psi} = gK\psi, \tag{14}$$

whose nonzero angular frequencies are g and $3g$. A second, reversible wave-type completion uses a pair of real three-component variables u, v:

$$\dot{u} = gK^{1/2}v, \qquad \dot{v} = -gK^{1/2}u, \tag{15}$$

where $K^{1/2}$ is the positive square root. Its nonzero frequencies are g and $\sqrt{3}g$. Equivalently, $\ddot{u} = -g^2Ku$. The positive norm $\|u\|^2+\|v\|^2$ is conserved. This wave realization is *not* graph-local in the same sense as incidence dynamics, because $K^{1/2}$ is generally dense.

A local incidence realization can instead be formulated on vertex and edge spaces. Choose an oriented incidence matrix $\partial : \mathbb{R}^{|E|} \to \mathbb{R}^{|V|}$ so that $K = \partial\partial^T$, and set

$$\mathbb{A} = \begin{pmatrix} 0 & g\partial \\ -g\partial^T & 0 \end{pmatrix}, \qquad \mathbb{A}^T = -\mathbb{A}, \qquad \mathbb{A}^2 = -g^2 \begin{pmatrix} K & 0 \\ 0 & \partial^T\partial \end{pmatrix}. \tag{16}$$

Thus $(\dot{u}, \dot{w}) = \mathbb{A}(u, w)$ conserves $\|u\|^2 + \|w\|^2$ and reproduces the nonzero wave frequencies $g, \sqrt{3}g$ on the vertex sector. Importantly, the identity $\mathbb{A}^2 = -g^2K$ is a *vertex-block identity*, not an equality between matrices acting on different spaces.

Proposition 5.3 (Kinematic nonuniqueness)**.** *The positive graph Laplacian K of Example 5.2, even together with exact conservation of a positive quadratic norm under closed evolution, does not select a unique nonzero frequency ratio.*

Proof. Equation Eq. (14) is unitary and has ratio $3/1 = 3$. The skew-adjoint flow Eq. (16) conserves its positive Euclidean norm and has ratio $\sqrt{3}/1 = \sqrt{3}$. The two constructions use the same K and fulfill the stated conservation requirement, but have different spectra. Thus those premises alone cannot select between them. □

The ratios are invariant under one common rescaling of the time parameter within each model. Calling a ratio *clock-free* means only that an overall clock calibration cancels; any comparison to experimental frequencies still requires a physical preparation/measurement correspondence. The preceding is a mathematical discriminator, not evidence that either completion describes nature.

6 A restricted signature obstruction

The relation between generated elliptic operators and possible Lorentzian descriptions should be stated with its assumptions visible.

Proposition 6.1 (Positive Schur complements preserve positivity)**.** *Let $M = \left(\begin{smallmatrix} A & B \\ B^T & C \end{smallmatrix}\right)$ be a real symmetric strictly positive-definite matrix, with C invertible. Then its Schur complement $A - BC^{-1}B^T$ is strictly positive-definite.*

Proof. For nonzero x, put $y = -C^{-1}B^Tx$. Since $(x, y) \neq 0$,

$$x^T(A - BC^{-1}B^T)x = \begin{pmatrix} x \\ y \end{pmatrix}^T M \begin{pmatrix} x \\ y \end{pmatrix} > 0. \tag{17}$$

□

Consequently, a positive-definite *finite-dimensional principal-symbol quadratic form*, reduced through a positive block Schur complement satisfying these hypotheses, cannot acquire mixed Lorentzian signature by that reduction alone. This does *not* prove an impossibility for every projection, dynamical extension, frequency-dependent elimination, singular limit, or nonlocal effective theory. In particular, the incidence evolution in Eq. (16) adds a dynamical sector; it is not a static signature change of the sort covered by Proposition 6.1. Whether a particular continuum HSMT model supports a Lorentzian effective description remains an independent question.

7 What a physical-selection principle must supply

The core does not determine an operational theory. We make that limitation constructive by listing distinct missing maps. First, a *generator-selection map* must choose the physical law of evolution from the admissible operator families. Second, a *calibration map* must relate its parameter to physical clock readings. Third, a *sector/observable map* must specify what laboratory preparations, observables, and outcomes correspond to mathematical states and operators. Fourth, a *geometric correspondence* must identify which of the possible response quantities behaves as measured distance or dimension.

An additional conservation axiom, considered in the HSMT program, states that closed fundamental evolution preserves a canonical positive quadratic form. Within a declared linear finite-dimensional model this restricts a generator G through $G^T M + MG = 0$ for some selected $M > 0$; however there are generally many such G. Thus conservation alone is insufficient for unique selection, as Proposition 5.3 illustrates concretely.

The broader program has considered causal-order, operational-tomography, information-preservation, and laboratory-emulation premises. These investigations can be informative, but no claim that *all possible* external physical premises have been exhausted follows from a finite set of audits. It is appropriate to suspend one line of foundational search without treating that suspension as a no-go theorem about every future physical law.

8 Relationship to the broader HSMT program

The research record motivating this article includes proposed finite hierarchy phases, a scalable relational geometry construction, ring-exchange analogues, and a laboratory comparator. Their relationship to this framework is summarized in Table 1. In this article the switch graph and its spectra are completely explicit; the reported hierarchy phases and their continuum limits are not. A future technical companion should state all finite generators and exact energies, give reproducible parameter domains, identify finite-size and limit orders, and prove the asserted convergence results. The absence of such content from this article must not be conflated with a proof that the content cannot exist.

9 Reproducibility and limitations

All numerical values presented as results *of this article* are finite exact quantities recoverable from Eq. (13). For example, the characteristic polynomial of K is $\lambda(\lambda-1)(\lambda-3)$; one may verify the results using symbolic arithmetic without external data. No new physical measurements,

Table 1: Claim ledger for the revised framework article. "Proved here" includes standard statements reproved with explicit assumptions; it does not imply novelty.

Statement	Status in this article
Minimal grading and bounded perturbation	Definition and established theorem
Exact block resolvent and projected memory	Established fact, derived under stated domains
Continuous parent coarse-graining	Established fact, proved here
Finite 2×2 switch graph and P_3 spectrum	Explicit reproducible benchmark
Distinct reversible generator frequency ratios	Conditional models; nonuniqueness proved
Positive Schur complement signature obstruction	Proved under strict positivity assumptions
Four-state BEGC minimality, $J(8,3)$ hierarchy	Program claims; independent proof not included
Scalable phase and transportation-polytope continuum	Program claims; independent proof not included
Unique physical grade, time, measurement, dimension	Open physical identifications
P5 hardware measurement or physical validation	Not claimed; no bench results provided

vendor SPICE simulations, or statistical datasets are claimed. A model-specific experiment comparing the two frequency ratios would test specified implementations, not independently establish HSMT as fundamental physics.

The principal limitation is intentional. This paper does not supply a novel physical law, a unique interpretation of the grade label, a derived $3 + 1$ spacetime, or a framework-level quantitative prediction. Nor does it provide complete proofs of the more ambitious hierarchy-generation and continuum claims summarized in Table 1. Those claims should not be submitted as self-standing theorems without accessible derivations. The mathematical contribution of the present article is therefore more modest than the entire program it describes.

10 Conclusion

HSMT can be formulated without conflating grading, spectral reduction, dimension measurement, and physical dynamics. Under explicit hypotheses its operator reductions are exact; a continuous parent can recover discrete grading; a finite contingency-switch system supplies a reproducible relational Laplacian; and two equally norm-preserving dynamic completions of the same graph yield distinct spectral ratios. A restricted Schur-complement argument explains why positive static reduction alone cannot provide an indefinite Lorentzian principal form, while leaving open other dynamical mechanisms.

The decisive unresolved issue is selection, not the mere availability of mathematical structures. Independent physical premises and operational correspondences would be needed to identify one dynamics and one experimentally meaningful interpretation. The reported BEGC and larger hierarchy phases are worthwhile targets for detailed companion publications, but their acceptance requires complete definitions and independently checkable proofs. This is a framework paper, not a claim that a physical theory has been uniquely derived.

Statements and declarations

Funding. No funding was received for this work.
Competing interests. The author declares no competing interests.

Data and code availability. No new experimental datasets or software packages are reported in this article. All finite benchmark matrices and exact spectra needed to reproduce its calculations are displayed in the text.

Generative AI assistance. OpenAI ChatGPT and xAI Grok were used as analytical and editorial assistants during drafting and review. Responsibility for verifying the mathematical and bibliographic content, approving the manuscript, and any eventual submission remains with the author.

References

[1] T. Kato, *Perturbation Theory for Linear Operators*, Springer (1966).

[2] M. Reed and B. Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, Academic Press (1972).

[3] H. Feshbach, "Unified theory of nuclear reactions," *Annals of Physics* **5**, 357–390 (1958).

[4] H. Feshbach, "A unified theory of nuclear reactions. II," *Annals of Physics* **19**, 287–313 (1962).

[5] V. Bach, T. Chen, J. Fröhlich, and I. M. Sigal, "Smooth Feshbach map and operator-theoretic renormalization group methods," *Journal of Functional Analysis* **203**, 44–92 (2003), doi:10.1016/S0022-1236(03)00057-0.

[6] P. Diaconis and B. Sturmfels, "Algebraic algorithms for sampling from conditional distributions," *Annals of Statistics* **26**, 363–397 (1998), doi:10.1214/aos/1030563990.

[7] F. R. K. Chung, *Spectral Graph Theory*, CBMS Regional Conference Series in Mathematics, Vol. 92, American Mathematical Society (1997), doi:10.1090/cbms/092.

[8] B. Sturmfels, *Gröbner Bases and Convex Polytopes*, University Lecture Series, Vol. 8, American Mathematical Society (1996).